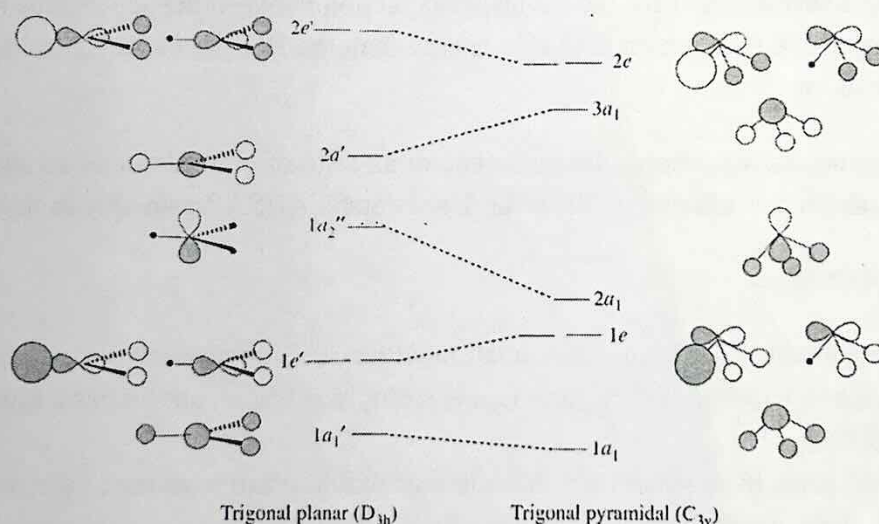


Quiz-2
Spring Semester 2024
General and Structural Chemistry
IIIT-Hyderabad

Max Marks: 50

Q1. What is meant by the *symmetry adapted linear combinations* (SALCs)? Construct it for the ground state of BeH_2 molecule. [2,3]

Q2. The Walsh correlation diagram for the *valence electrons* of an AH_3 molecules is shown below. The left side of the diagram gives the energy ordering (increasing from bottom to top) of the molecular orbitals for an H-A-H bond angle for a trigonal planar geometry. The right side gives the energy ordering of the molecular orbitals for an H-A-H bond angle for a trigonal pyramidal geometry.



Based on the above Walsh correlation diagram, answer the following:

[1,1,1,3,1]

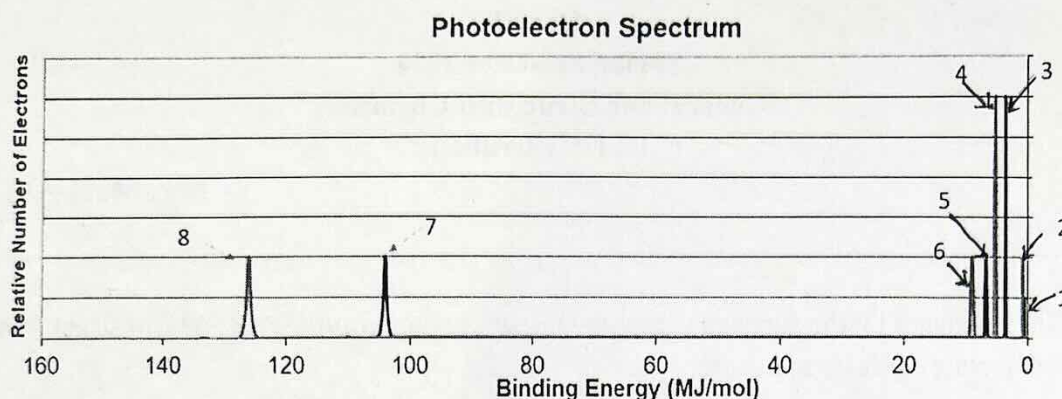
- (a) What is the expected H-C-H bond angle in CH_3^+ ion?
- (b) Give the MO electron distribution for the ground state CH_3^+ ion.
- (c) What are the HOMO and LUMO of the CH_3^+ ion?
- (d) When CH_3^+ ion shows its Lewis Acid character, which atoms act as the electron pair acceptor? Why? What will be the geometry of the Lewis Acid-Base adduct? Why?
- (e) While changing trigonal planar to trigonal pyramidal geometry, which orbitals get stabilized, and which get destabilized?

Q3. (a) Compare PES and ESCA techniques.

[3]

(b) PES line/peaks of Na and Mg are shown below. Identify the peaks (1 – 8) with corresponding element's AOs.

[4]



Q4. When two molecules come into contact, there is always a possibility of bonding interactions between the two, since each molecule has filled MO's and vacant MO's. Explain why sometimes they form adducts and sometime not. [3]

Q5. Show the FMO diagram for the covalent interaction between the acceptor A and donor B when the donor HOMO is much higher in energy than the HOMO of the acceptor. Comment on the nature of interactions. [3,3]

Q6. Some metals are commonly found in nature as salts of chloride or as oxide ores while others are found in combination with sulfur. Use Pearson's HSAB principle to justify it. [4]

Q7. State true or false: [4×2]

- (i) In general, a hard acid has a small ionic radius, high positive charge, no electron pairs in its valence shell, low EA, is strongly solvated, and/or has a high-energy HOMO.
- (ii) Soft acids have small radii, have low or partial positive charge, have electron pairs in their valence shells, are easy to polarize and oxidize, and have low-energy LUMOs with large magnitude coefficients.
- (iii) Hard bases have small radii, are hard to oxidize, are weakly polarizable, have very electronegative centers, are strongly solvated, and/or have low-energy HOMOs.
- (iv) Soft bases tend to have large radii, have intermediate values for the electronegativity, are easily polarized and oxidized, and have low-energy HOMOs with large magnitude coefficients.

Q8. Define the valence and conduction bands. What are the conduction bands in Na and Mg metals respectively. [2,2]

Q9. State the rules for orbital hybridization. [4]

Q10. Calculate the bond angle on the central atom which undergoes sp^2 hybridization. [2]